In [ ]:
```
import numpy as np
import pandas as pd


import matplotlib.pyplot as plt
import seaborn as sns
pd.set_option('display.float_format', lambda x: '%.3f' % x)
```

In [ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to
forcibly remount, call drive.mount("/content/drive",
force_remount=True).
```

In [ ]:
```
df = pd.read_csv('/content/foodhub_order.csv')
```

In [ ]:
```
data.head()
```

	order_id	customer_id	restaurant_name	cuisine_type	cost_
0	1477147	337525	Hangawi	Korean	30.75
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08
2	1477070	66393	Cafe Habana	Mexican	12.23
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20
4	1478249	76942	Dirty Bird to Go	American	11.59

In [ ]:
```
data.tail()
```

	order_id	customer_id	restaurant_name	cuisine_type
1893	1476701	292602	Chipotle Mexican Grill $1.99 Delivery	Mexican
1894	1477421	397537	The Smile	American
1895	1477819	35309	Blue Ribbon Sushi	Japanese
1896	1477513	64151	Jack's Wife Freda	Mediterranean
1897	1478056	120353	Blue Ribbon Sushi	Japanese

In [ ]:
```
print(data.shape[0], "raws and", data.shape[1], "columns")
```

```
1898 raws and 9 columns
```

In [ ]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   order_id               1898 non-null   int64  
 1   customer_id            1898 non-null   int64  
 2   restaurant_name        1898 non-null   object 
 3   cuisine_type           1898 non-null   object 
 4   cost_of_the_order      1898 non-null   float64
 5   day_of_the_week        1898 non-null   object 
 6   rating                 1898 non-null   object 
 7   food_preparation_time  1898 non-null   int64  
 8   delivery_time          1898 non-null   int64  
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```

In [ ]:
```
data['rating'].unique()
```

```
array(['Not given', '5', '3', '4'], dtype=object)
```

In [19]:
```
df.isnull().sum()
```

Out[19]:

	0
order_id	0
customer_id	0
restaurant_name	0
cuisine_type	0
cost_of_the_order	0
day_of_the_week	0
rating	736
food_preparation_time	0
delivery_time	0

dtype: int64

In [29]:
```
df.describe()
```

Out[29]:

	order_id	customer_id	cost_of_the_order	rating
count	1898.000	1898.000	1898.000	1162.000
mean	1477495.500	171168.478	16.499	4.344
std	548.050	113698.140	7.484	0.741
min	1476547.000	1311.000	4.470	3.000
25%	1477021.250	77787.750	12.080	4.000
50%	1477495.500	128600.000	14.140	5.000
75%	1477969.750	270525.000	22.297	5.000
max	1478444.000	405334.000	35.410	5.000

In [25]:
```
df['rating'].value_counts()
```

Out[25]:

	count
rating	
5.000	588
4.000	386
3.000	188

dtype: int64

In [31]:
```
df['rating'].isnull().sum()
```

Out[31]: np.int64(736)

In [32]:
```
df['order_id'].nunique()
```

Out[32]: 1898

In [33]:
```
df['customer_id'].nunique()
```

Out[33]: 1200

In [34]:
```
df['restaurant_name'].nunique()
```

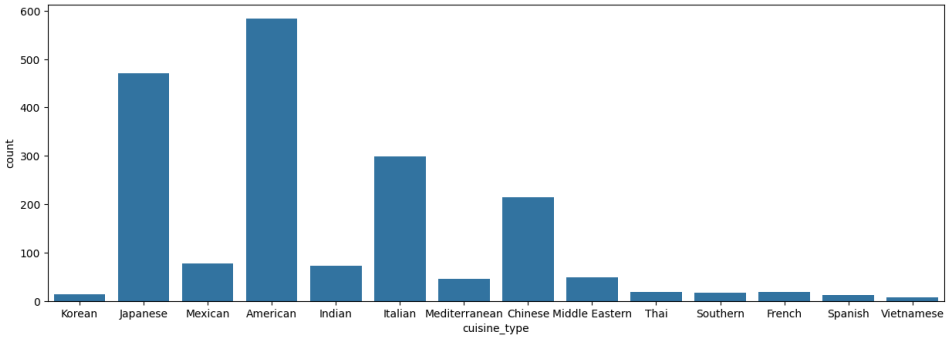
Out[34]: 178

In [35]:
```
df['cuisine_type'].nunique()
```

Out[35]: 14

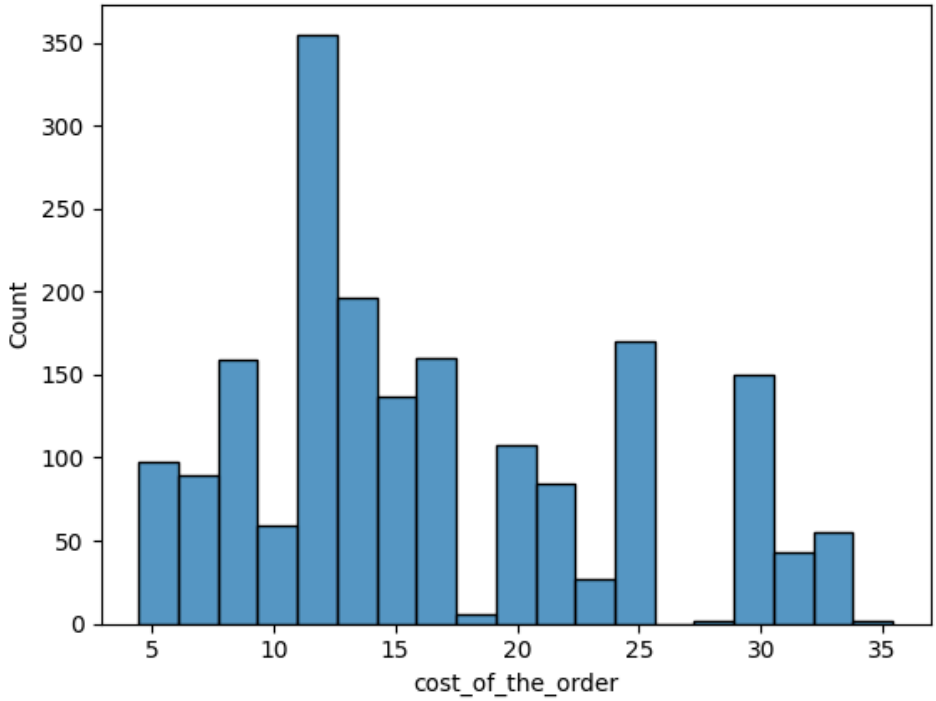
Cuisine Type Distribution: American (600) and Japanese (470) cuisines are the most ordered, while cuisines like Vietnamese, Spanish, French, and Thai see very low demand (under 25 orders each), showing a strong customer preference toward mainstream

In [36]:
```
plt.figure(figsize=(15,5))
sns.countplot(data=df, x='cuisine_type')
plt.show()
```



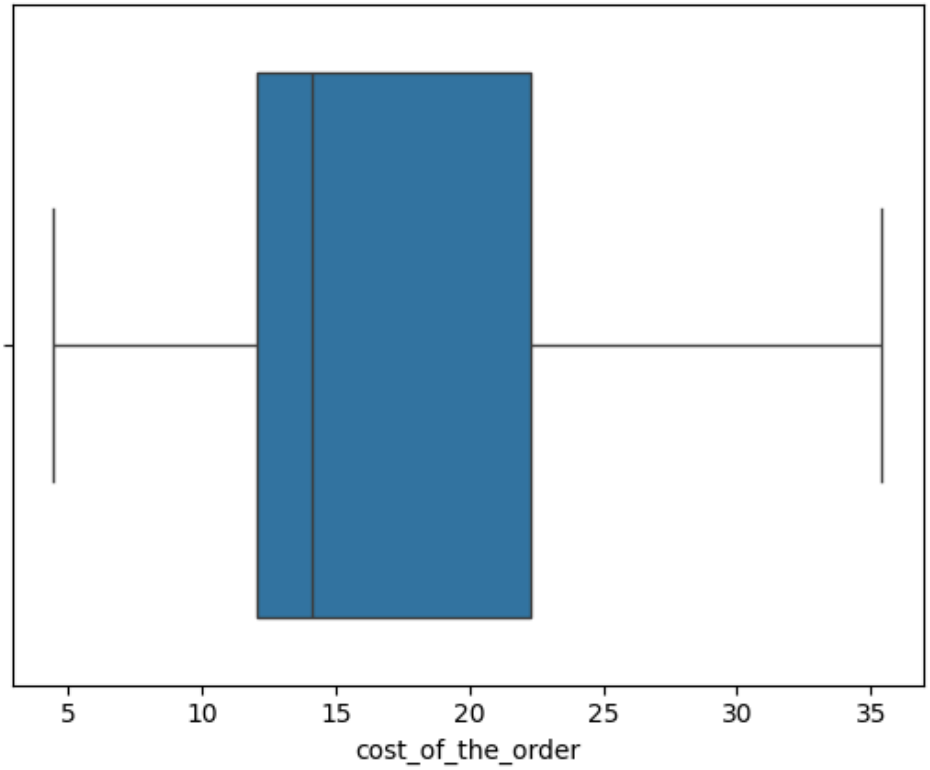
Cost of the Order Distribution: Most orders are priced between ₹10–₹25, with a sharp peak around ₹12 (355 orders), while very few orders fall below ₹5 or above ₹30 indicating customers typically spend in the mid-range rather than on very cheap or very expensive orders.

In [37]:
```
sns.histplot(data=df, x='cost_of_the_order')
plt.show()
```



The median order cost is around ₹14, with the middle 50% of orders falling between ₹12–₹22, and no outliers present — showing the cost distribution is fairly symmetric with no extreme pricing.bold text

In [38]:
```
sns.boxplot(data=df, x='cost_of_the_order')
plt.show()
```

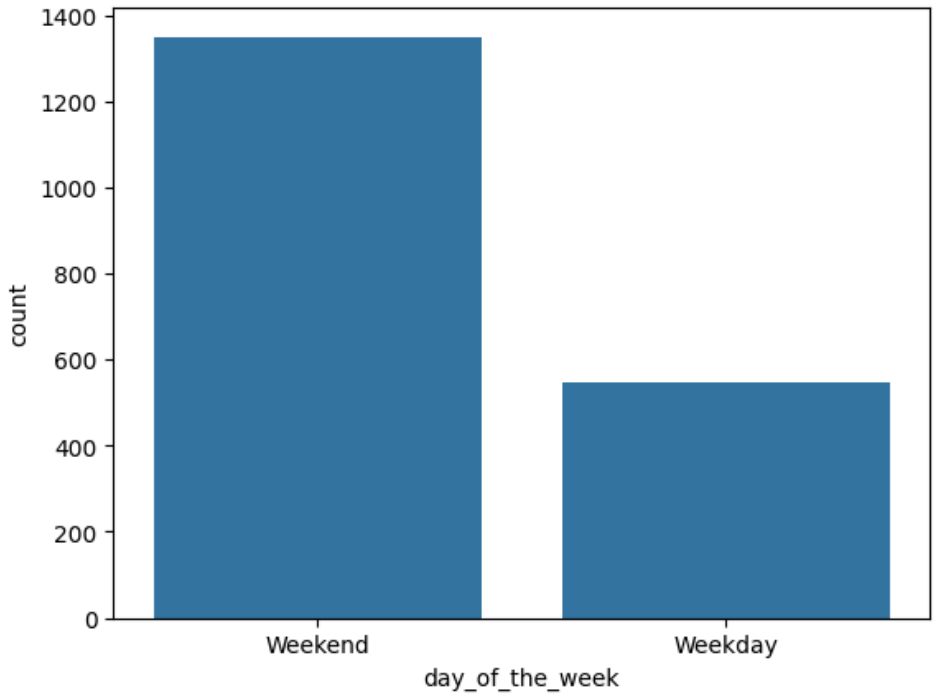


In [39]:
```
df['day_of_the_week'].unique()
```

Out[39]:
```
array(['Weekend', 'Weekday'], dtype=object)
```

Orders are significantly higher on weekends (1350) compared to weekdays (550), indicating customers prefer ordering food more on weekends, likely due to more free time or social occasions.

In [40]:
```
sns.countplot(data=df, x='day_of_the_week')
plt.show()
```

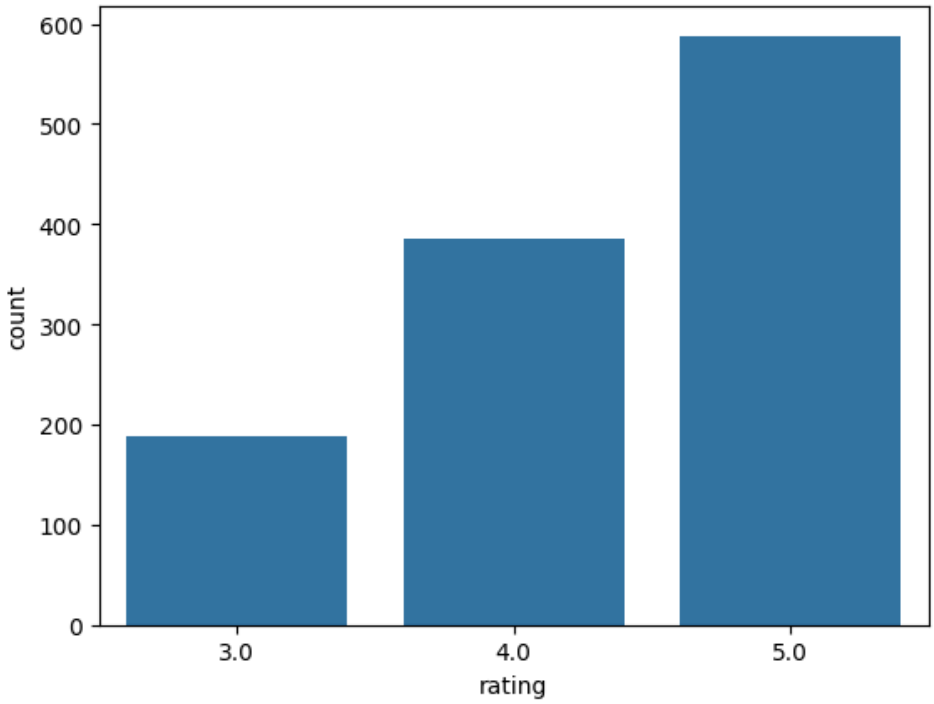


In [41]:
```
df['rating'].unique()
```

Out[41]:
```
array([nan,  5.,  3.,  4.])
```

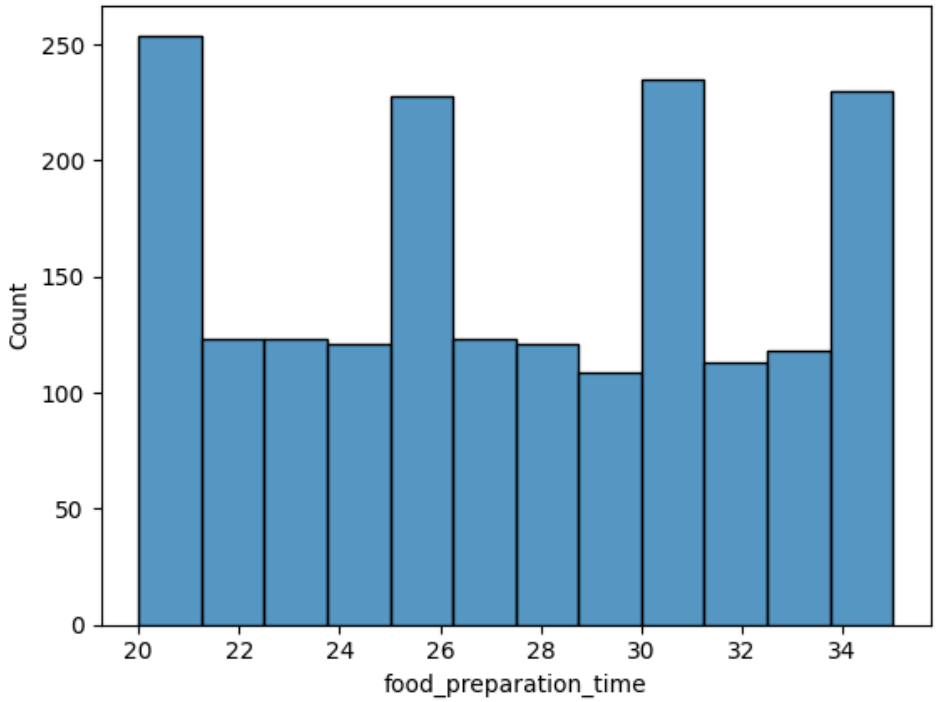
Most customers give high ratings, with 5.0 being the most common (590 orders) followed by 4.0 (385), while very few give a rating of 3.0 (190) showing overall strong customer satisfaction.

In [42]:
```
sns.countplot(data=df, x='rating')
plt.show()
```



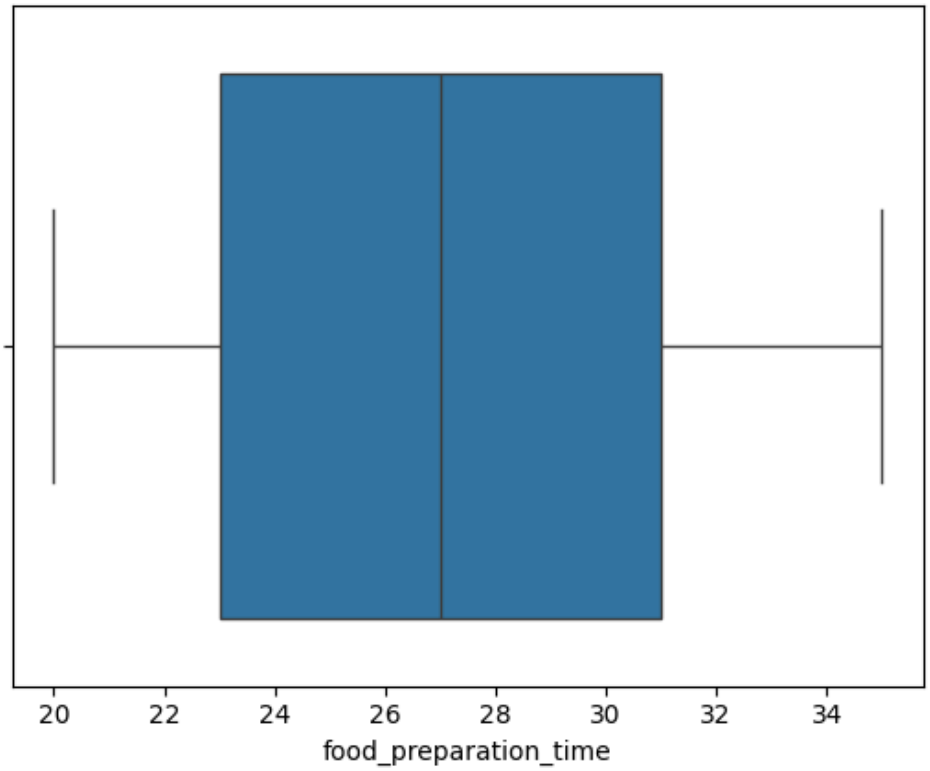
Preparation time is fairly evenly spread between 20–35 minutes, with no strong peak, suggesting food prep time is fairly consistent across orders regardless of cuisine or cost.

In [43]:
```
sns.histplot(data=df, x='food_preparation_time')
plt.show()
```



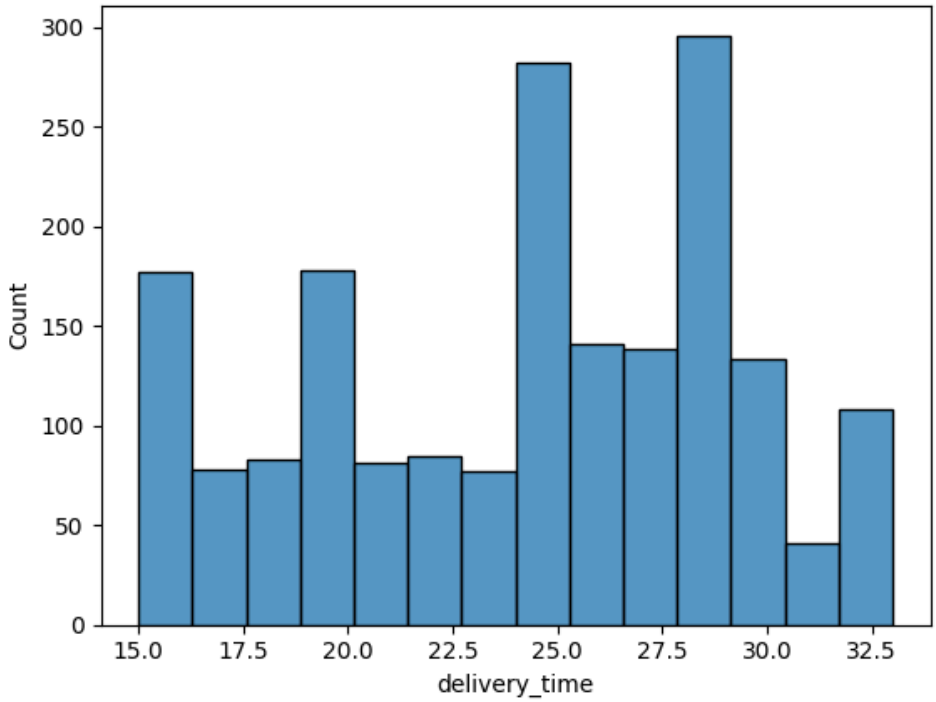
The median preparation time is around 27 minutes, with most orders falling between 23–31 minutes and no outliers, indicating a stable and predictable food preparation process. as same as histplot says

In [44]:
```
sns.boxplot(data=df, x='food_preparation_time')
plt.show()
```



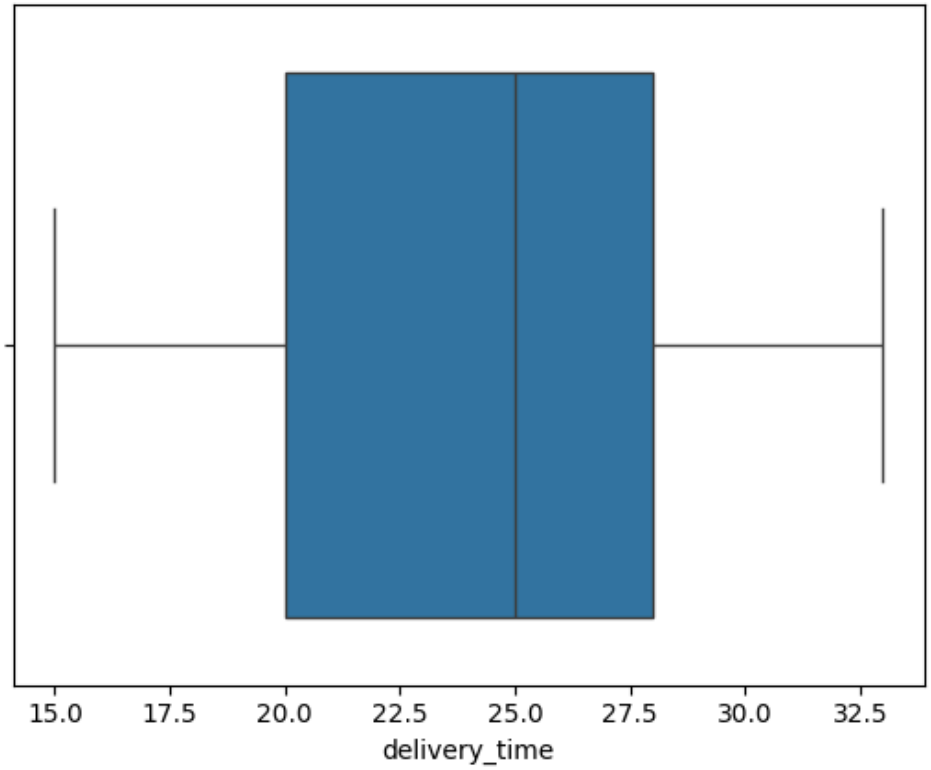
Delivery times are concentrated between 24–29 minutes, with peaks around 25 and 28 minutes, showing that most deliveries take a bit under half an hour.

In [45]:
```
sns.histplot(data=df, x='delivery_time')
plt.show()
```



The median delivery time is around 25 minutes, with most deliveries falling between 20–28 minutes and no outliers, indicating a fairly consistent delivery performance.

In [46]:
```
sns.boxplot(data=df, x='delivery_time')
plt.show()
```



Shake Shack is the most ordered restaurant by far (219 orders), followed by The Meatball Shop (132) and Blue Ribbon Sushi (119), while Blue Ribbon Fried Chicken (96) and Parm (68) complete the top 5 showing a small set of restaurants drive a large share of total orders.

In [47]:
```
df['restaurant_name'].value_counts().head(5)
```

Out[47]:

	count
restaurant_name	
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

dtype: int64

American cuisine leads weekend orders (415), followed by Japanese (335) and Italian (207), while cuisines like Vietnamese (4), Korean, Southern, and Spanish (11 each) see minimal weekend demand confirming American and Japanese remain top choices even on weekends.

In [48]:
```
df_weekend = df[df['day_of_the_week'] == 'Weekend']
df_weekend['cuisine_type'].value_counts()
```

Out[48]:

	count
cuisine_type	
American	415
Japanese	335
Italian	207
Chinese	163
Mexican	53
Indian	49
Middle Eastern	32
Mediterranean	32
Thai	15
French	13
Korean	11
Southern	11
Spanish	11
Vietnamese	4

dtype: int64

In [49]:
```
df_greater_than_20 = df[df['cost_of_the_order'] > 20]
print('The number of total orders that cost above 20 dollars
is:',
      df_greater_than_20.shape[0])
percentage = (df_greater_than_20.shape[0] / df.shape[0]) * 100

print("Percentage of orders above 20 dollars:", round(percentage,
2), '%')
```

```
The number of total orders that cost above 20 dollars is: 555
Percentage of orders above 20 dollars: 29.24 %
```

In [52]:
```
mean_del_time = df['delivery_time'].mean()
print('The mean delivery time for this dataset is',
round(mean_del_time, 2), 'minutes')
```

```
The mean delivery time for this dataset is 24.16 minutes
```

customer IDs and number of orders placed by the top 3 customers.

In [53]:
```
df['customer_id'].value_counts().head(3)
```

Out[53]:

	count
customer_id	
52832	13
47440	10
83287	9

dtype: int64

Southern and Middle Eastern cuisines tend to have higher median order costs (16–17), while Korean and Vietnamese have the lowest medians (12) and tighter cost ranges with Korean and Mediterranean showing a few outliers, indicating occasional high-cost orders in those categories.

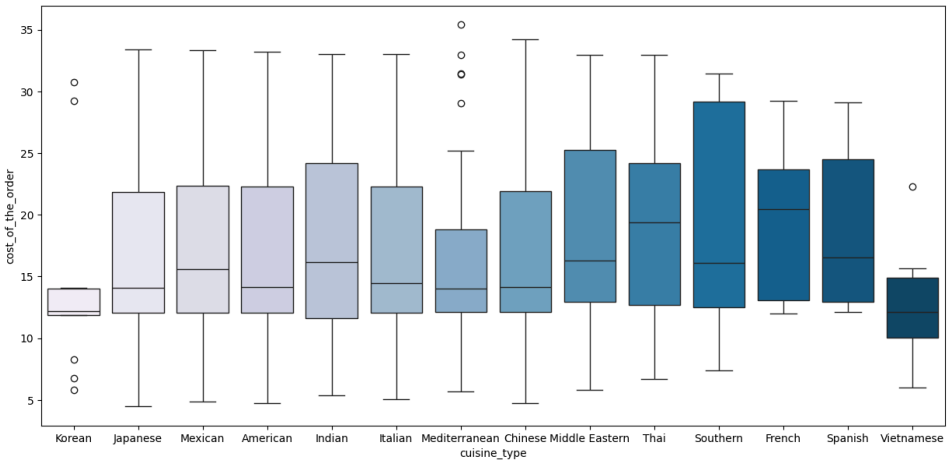
In [56]:
```
plt.figure(figsize=(15,7))
sns.boxplot(x='cuisine_type',
y='cost_of_the_order',
data=df,
palette='PuBu')

plt.show()
```

```
/tmp/ipykernel_12440/607064422.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated
and will be removed in v0.14.0. Assign the `x` variable
to `hue` and set `legend=False` for the same effect.

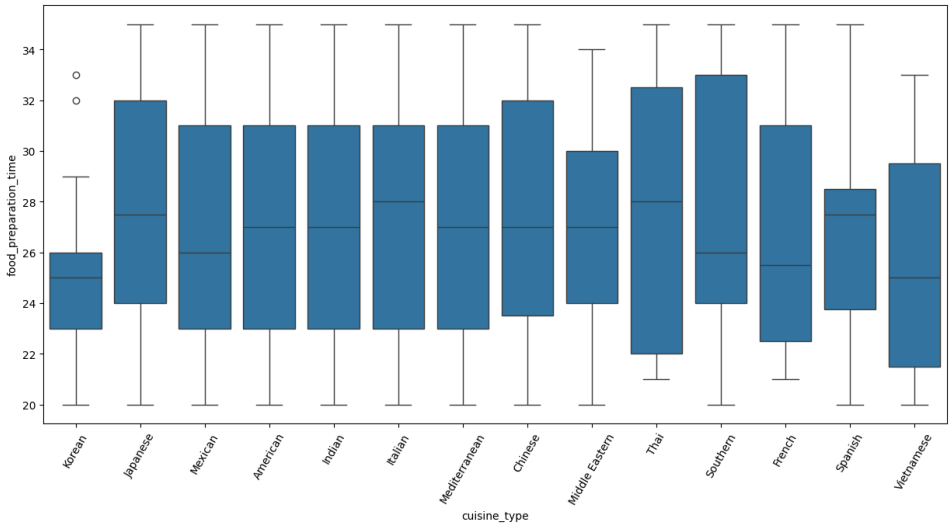
  sns.boxplot(x='cuisine_type',
```



Cuisine vs Food Preparation Time: Korean cuisine has the lowest median preparation time around 25 minutes with two outliers on the higher side while most other cuisines like Thai and Southern have higher medians around 26 to 28 minutes with wide ranges showing preparation time varies noticeably across cuisines

In [57]:
```
plt.figure(figsize=(15,7))
sns.boxplot(x='cuisine_type',
y='food_preparation_time',
data=df)

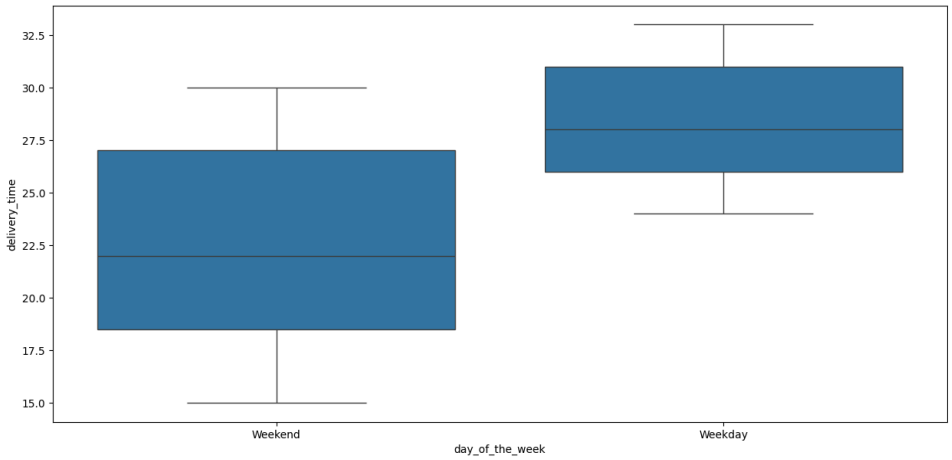
plt.xticks(rotation=60)
plt.show()
```



Day of the Week vs Delivery Time: Weekday orders have a noticeably higher median delivery time around 28 minutes compared to weekends where the median is around 22 minutes suggesting deliveries are slower on weekdays likely due to traffic or staffing differences

In [58]:
```
plt.figure(figsize=(15,7))
sns.boxplot(x='day_of_the_week',
y='delivery_time',
data=df)

plt.show()
```



Revenue Generated by Restaurants: Shake Shack generates the highest revenue by far at 3579.53 followed by The Meatball Shop at 2145.21 and Blue Ribbon Sushi at 1903.95 while the remaining restaurants show a steady decline down to Five Guys Burgers and Fries at 506.47 confirming Shake Shack as the clear revenue leader

In [60]:
```
df.groupby(['restaurant_name'])
['cost_of_the_order'].sum().sort_values(
ascending=False
).head(14)
```

Out[60]:

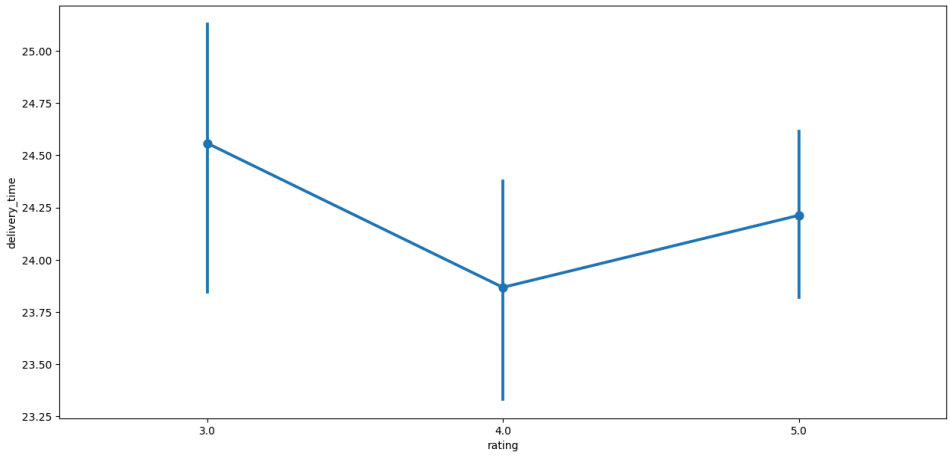
	cost_of_the_order
restaurant_name	
Shake Shack	3579.530
The Meatball Shop	2145.210
Blue Ribbon Sushi	1903.950
Blue Ribbon Fried Chicken	1662.290
Parm	1112.760
RedFarm Broadway	965.130
RedFarm Hudson	921.210
TAO	834.500
Han Dynasty	755.290
Blue Ribbon Sushi Bar & Grill	666.620
Rubirosa	660.450
Sushi of Gari 46	640.870
Nobu Next Door	623.670
Five Guys Burgers and Fries	506.470

dtype: float64

Rating vs Delivery Time: Orders with a rating of 3.0 have the highest average delivery time around 24.5 minutes while ratings of 4.0 have the lowest average around 23.9 minutes and rating of 5.0 sits around 24.2 minutes suggesting delivery time alone does not strongly determine customer rating

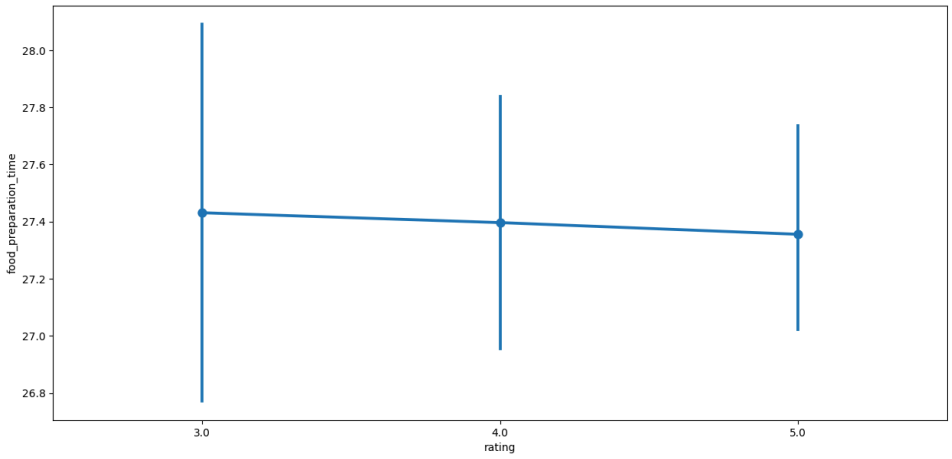
In [61]:
```
plt.figure(figsize=(15, 7))
sns.pointplot(x='rating',
y='delivery_time',
data=df)

plt.show()
```



Rating vs Food Preparation Time: Average food preparation time stays almost the same across all ratings around 27.3 to 27.4 minutes indicating food preparation time has little to no impact on the rating customers give

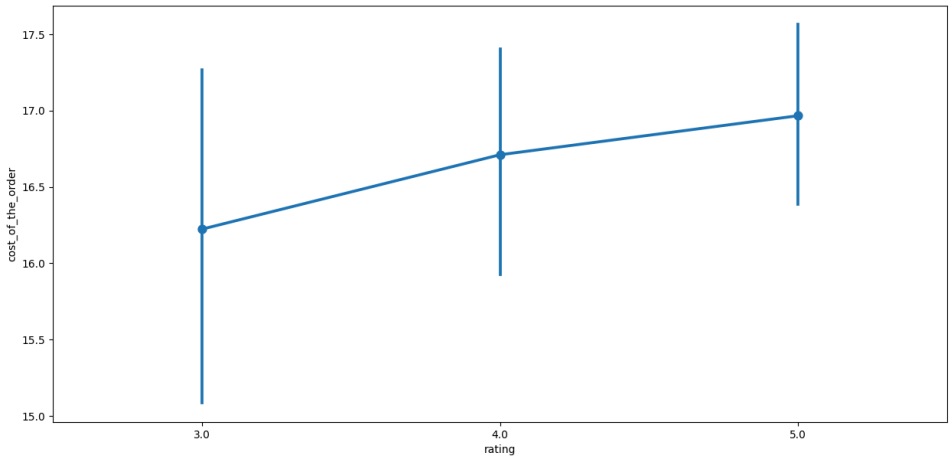
In [62]:
```
plt.figure(figsize=(15, 7))
sns.pointplot(x='rating',
y='food_preparation_time',
data=df)
plt.show()
```



Rating vs Cost of the Order: Average order cost increases slightly as rating increases from about 16.2 for rating 3.0 to about 17.0 for rating 5.0 suggesting customers ordering higher priced items tend to give slightly better ratings

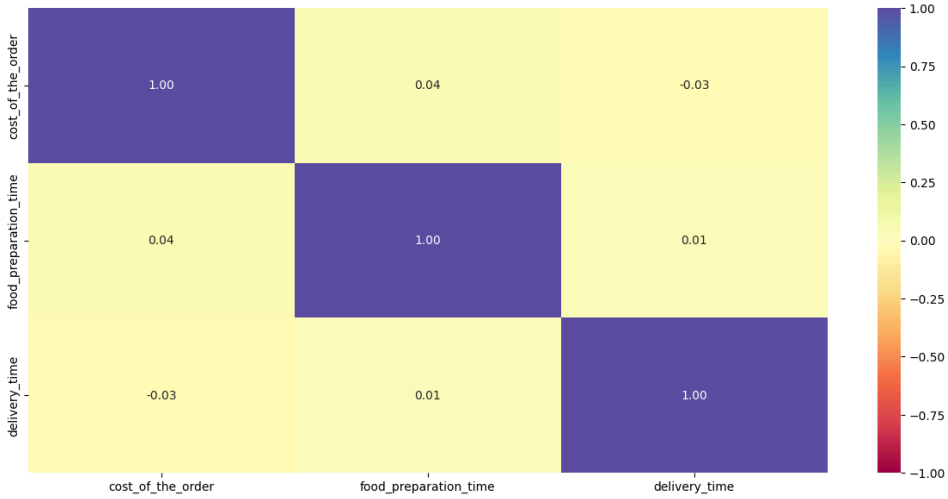
In [64]:
```
plt.figure(figsize=(15, 7))
sns.pointplot(x='rating',
y='cost_of_the_order',
data=df)

plt.show()
```



Insights: Since cost of the order, food preparation time, and delivery time show almost no correlation with each other, it means these three operational factors work independently in this business. A costlier order does not take longer to prepare or deliver, food preparation time does not influence delivery time, and delivery time is not linked to how expensive the order is. This suggests delays or inefficiencies in one process (like food prep) are not caused by or connected to the other processes, so each stage (pricing, kitchen operations, delivery logistics) should be optimized separately rather than assuming improvements in one will affect another.

In [66]:
```
col_list =
['cost_of_the_order','food_preparation_time','delivery_time']
plt.figure(figsize=(15, 7))
sns.heatmap(df[col_list].corr(),
annot=True,
vmin=-1,
vmax=1,
fmt=".2f",
cmap="Spectral")
plt.show()
```



In [69]:
```
df_rating_count = df_rated.groupby(['restaurant_name'])
['rating'].count().sort_values(ascending=False).reset_index()
df_rating_count.head()
```

Out[69]:

	restaurant_name	rating
0	Shake Shack	133
1	The Meatball Shop	84
2	Blue Ribbon Sushi	73
3	Blue Ribbon Fried Chicken	64
4	RedFarm Broadway	41

In [71]:
```
rest_names = df_rating_count[df_rating_count['rating'] > 50]
['restaurant_name']
df_mean_4 =
df_rated[df_rated['restaurant_name'].isin(rest_names)].copy()
```

In [74]:
```
# mean rating of each restaurant
df_mean_4.groupby(['restaurant_name'])
['rating'].mean().sort_values(ascending=False).reset_index().drop
na()
```

Out[74]:

	restaurant_name	rating
0	The Meatball Shop	4.512
1	Blue Ribbon Fried Chicken	4.328
2	Shake Shack	4.278
3	Blue Ribbon Sushi	4.219

In [75]:
```
# restaurants with an average rating greater than 4
restaurant_rating = df_mean_4.groupby('restaurant_name')
['rating'].agg(['count', 'mean']).reset_index()
restaurant_rating[(restaurant_rating['count'] > 50) &
(restaurant_rating['mean'] > 4)]
```

Out[75]:

	restaurant_name	count	mean
0	Blue Ribbon Fried Chicken	64	4.328
1	Blue Ribbon Sushi	73	4.219
2	Shake Shack	133	4.278
3	The Meatball Shop	84	4.512

In [78]:
```
def compute_rev(x):
    if x > 20:
        return x * 0.25
    elif x > 5:
        return x * 0.15
    else:
        return x * 0
```

In [80]:
```
df['Revenue'] = df['cost_of_the_order'].apply(compute_rev)
df.head()
```

Out[80]:

	order_id	customer_id	restaurant_name	cuisine_type	cost_
0	1477147	337525	Hangawi	Korean	30.75
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08
2	1477070	66393	Cafe Habana	Mexican	12.23
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20
4	1478249	76942	Dirty Bird to Go	American	11.59

In [81]:
```
total_rev = df['Revenue'].sum()
print('The net revenue is around',
round(total_rev, 2),'dollars')
```

```
The net revenue is around 6166.3 dollars
```

In [82]:
```
df['total_time'] = df['food_preparation_time'] +
df['delivery_time']
```

In [83]:
```
df_more_than_60 = df[df['total_time'] > 60]
percentage = (df_more_than_60.shape[0] / df.shape[0]) * 100
print("Percentage of orders taking more than 60
minutes:",round(percentage, 2), '%')
```

```
Percentage of orders taking more than 60 minutes: 10.54 %
```

In [84]:
```
print('The mean delivery time on weekdays is around',round(
df[df['day_of_the_week'] == 'Weekday']
['delivery_time'].mean()),'minutes')
```

```
The mean delivery time on weekdays is around 28 minutes
```

In [85]:
```
print('The mean delivery time on weekends is around',round
(df[df['day_of_the_week'] == 'Weekend']
['delivery_time'].mean()),'minutes')
```

```
The mean delivery time on weekends is around 22 minutes
```

In [ ]:
```

```

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